

GROUP BY

Aula 06

Configuração

```
library(DBI)
library(RSQLite)
con <- dbConnect(RSQLite::SQLite(), "datasets/me315.sqlite")
```

Questão 1

```
SELECT STATE, MIN(ACRES) AS MINACRES
FROM PRESERVE
GROUP BY STATE;
```

STATE	MINACRES
AZ	380
MA	4
MT	121

Questão 2

```
SELECT STATE, MAX(ACRES) AS MAXACRES
FROM PRESERVE
WHERE ACRES <= 10000
GROUP BY STATE;
```

STATE	MAXACRES
AZ	1200
MA	830
MT	1130

Questão 3

```
SELECT STATE, MAX(ACRES) AS MAXACRES
FROM PRESERVE
WHERE ACRES <= 10000
GROUP BY STATE
ORDER BY MAXACRES DESC;
```

STATE	MAXACRES
AZ	1200
MT	1130

STATE	MAXACRES
MA	830

Questão 4

```
SELECT STATE, MAX(ACRES) AS MAXACRES
FROM PRESERVE
GROUP BY STATE
HAVING MAXACRES < 25000;
```

STATE	MAXACRES
MA	830
MT	15000

Questão 5

```
SELECT STATE, MIN(ACRES) AS MINACRES
FROM PRESERVE
GROUP BY STATE
HAVING MINACRES < 100;
```

STATE	MINACRES
MA	4

Questão 6

```
SELECT STATE, PNAME, MAX(ACRES) AS MAXACRES
FROM PRESERVE
WHERE ACRES > 1000
GROUP BY STATE
HAVING MAXACRES < 25000;
```

STATE	PNAME	MAXACRES
MT	PINE BUTTE SWAMP	15000

Questão 7

```
SELECT STATE, SUM(ACRES) AS TOTAL
FROM PRESERVE
GROUP BY STATE
HAVING TOTAL BETWEEN 10000 AND 50000;
```

STATE	TOTAL
MT	16931

Questão 8

```
SELECT STATE, AVG(ACRES) AS MEDIA
FROM PRESERVE
GROUP BY STATE
ORDER BY MEDIA DESC
LIMIT 1;
```

STATE	MEDIA
AZ	12840

Questão 9

```
SELECT DNO, AVG(SALARY) AS MEDIA
FROM EMPLOYEE
GROUP BY DNO;
```

DNO	MEDIA
10	1200.000
20	4666.667
40	500.000

Questão 10

```
SELECT DNO, SUM(SALARY) AS SOMA, MAX(SALARY) AS MAXIMO, MIN(SALARY) AS MINIMO
FROM EMPLOYEE
GROUP BY DNO;
```

DNO	SOMA	MAXIMO	MINIMO
10	2400	2000	400
20	14000	9000	2000
40	500	500	500

Questão 11

```
SELECT DNO, COUNT(*) AS NUM
FROM EMPLOYEE
WHERE DNO <> 40
GROUP BY DNO;
```

DNO	NUM
10	2
20	3

Questão 12

```
SELECT DNO, COUNT(*) AS NUM
FROM EMPLOYEE
GROUP BY DNO
HAVING NUM > 2;
```

DNO	NUM
20	3

Questão 13

```
SELECT STATE, SUM(ACRES) AS "SUM(ACRES)"
FROM PRESERVE
GROUP BY STATE
UNION ALL
SELECT 'Total', SUM(ACRES)
FROM PRESERVE;
```

STATE	SUM(ACRES)
AZ	51360
MA	1760
MT	16931
Total	70051

```
dbDisconnect(con)
```